Lesson 5.10 - Probability Tree Diagrams

> **Video:** Probability Tree Diagrams by The Organic Chemistry Tutor **Chapter:** 5 - Introduction to Probability

How a tree diagram works

- Each **branch** shows one possible outcome of a step, labeled with its probability.
- Branches leaving the same point **add up to 1**.
- **Along a path ("and")** → **multiply** the probabilities.
- **Combining paths ("or")** → **add** the path probabilities.
- All the final path probabilities add up to 1.

Problem 1 - with replacement (independent)

A bag holds **8 red** and **12 blue** marbles (20 total). Two marbles are drawn **with replacement**.

$P(R) = \frac{\text{red}}{\text{total}} = \frac{8}{20} = \frac{2}{5}$, $P(B) = \frac{\text{blue}}{\text{total}} = \frac{12}{20} = \frac{3}{5}$. These stay the same on every draw.

```
                2/5   R  →  RR : 2/5 · 2/5 = 4/25
        R   <
  2/5 /         3/5   B  →  RB : 2/5 · 3/5 = 6/25
start
  3/5 \         2/5   R  →  BR : 3/5 · 2/5 = 6/25
        B   <
                3/5   B  →  BB : 3/5 · 3/5 = 9/25
```

Check: $4 + 6 + 6 + 9 = 25$, so the total is $25/25 = 1$ ✓

(a) P(two blue) $= \frac{9}{25} = 0.36 = 36\%$

(b) P(at least one red): paths RR, RB, BR

$$P(RR) + P(RB) + P(BR) = \frac{4 + 6 + 6}{25} = \frac{16}{25} = 0.64$$

Complement shortcut: "at least one red" = "not both blue":

$$1 - P(BB) = 1 - 0.36 = 0.64 \checkmark$$

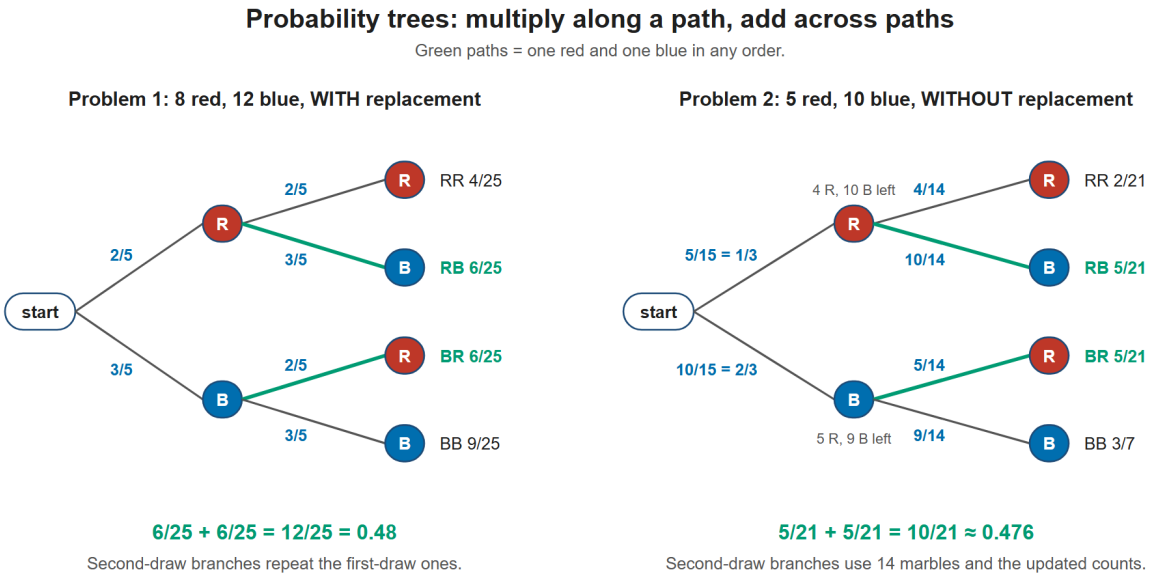
(c) P(red first, then blue): only RB

$$P(RB) = P(R) \cdot P(B) = \frac{2}{5} \cdot \frac{3}{5} = \frac{6}{25} = 0.24 = 24\%$$

(d) P(one red and one blue), in any order: RB or BR

$$P(RB) + P(BR) = \frac{6}{25} + \frac{6}{25} = \frac{12}{25} = 0.48 = 48\%$$

> (c) and (d) differ: "red **then** blue" fixes the order, while "a red **and** a blue" allows either order.



Multiply along each path to get its probability, then add the paths that match the question. Without replacement, the second-draw branches change depending on the first draw.

Problem 2 - without replacement (dependent)

A bag holds **5 red** and **10 blue** marbles (15 total). Sally draws two **without replacement**, so the second-draw probabilities depend on the first draw.

```
                4/14 = 2/7   R   →  RR : 1/3 · 2/7   = 2/21
        R  <                        (4 R, 10 B left)
  1/3   /       10/14 = 5/7  B   →  RB : 1/3 · 5/7   = 5/21
start
  2/3   \       5/14         R   →  BR : 2/3 · 5/14  = 5/21
        B  <                        (5 R, 9 B left)
                9/14         B   →  BB : 2/3 · 9/14  = 3/7
```

(a) P(two red)

$$P(RR) = P(R_1) \cdot P(R_2 \mid R_1) = \frac{1}{3} \cdot \frac{2}{7} = \frac{2}{21} \approx 0.095 = 9.5\%$$

(b) P(one red and one blue), in any order

$$P(RB) + P(BR) = \frac{5}{21} + \frac{5}{21} = \frac{10}{21} \approx 0.476 = 47.6\%$$

Problem 3 - three draws with replacement

3 blue, **7 red** (10 total). Megan draws 3 **with replacement**: $P(R) = 0.7$, $P(B) = 0.3$ every time.

There are $2^3 = 8$ outcomes: RRR, RRB, RBR, RBB, BRR, BRB, BBR, BBB.

(a) P(at least one red): every outcome except BBB

$$1 - P(BBB) = 1 - (0.3)^3 = 1 - 0.027 = 0.973 = 97.3\%$$

(b) P(at most two blue): 0, 1 or 2 blue, again everything except BBB

$$1 - P(BBB) = 1 - 0.027 = 0.973$$

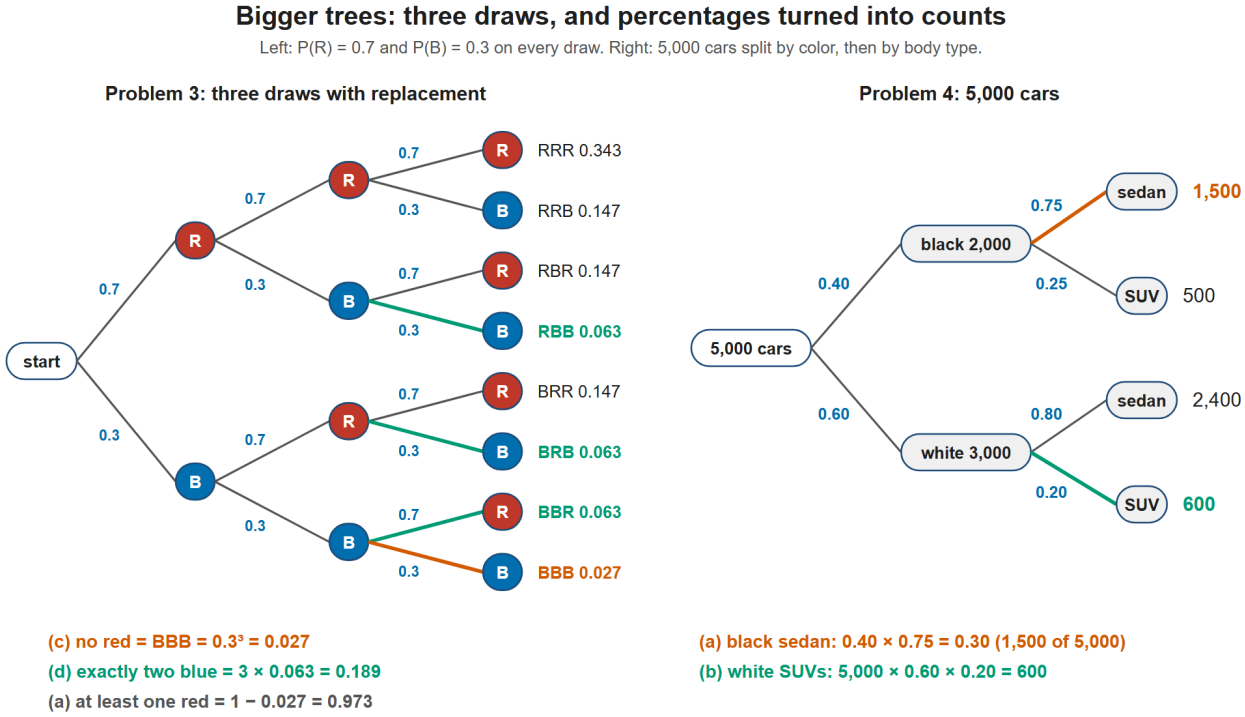
"At least one red" and "at most two blue" describe the **same** event.

(c) P(no red) = P(BBB)

$$P(BBB) = P(B)^3 = (0.3)^3 = 0.027 = 2.7\%$$

(d) P(exactly two blue): RBB, BRB, BBR

$$3 \times P(R)\,P(B)^2 = 3 \times (0.7)(0.3)(0.3) = 3 \times 0.063 = 0.189 = 18.9\%$$



Left: the 8 outcomes of three draws. Exactly two blue is three separate paths. Right: multiplying the counts by the branch percentages turns the tree into counts.

Problem 4 - percentages on a tree

There are **5,000** cars in a neighborhood. **40%** are black and **60%** are white. **75%** of the black cars are sedans (the rest SUVs). **20%** of the white cars are SUVs (the rest sedans).

```
                      0.75  sedan   1,500
        black  2,000 <
  0.40 /              0.25  SUV       500
5,000
  0.60 \              0.80  sedan   2,400
        white  3,000 <
                      0.20  SUV       600
```

(a) P(black sedan)

$$P(\text{black and sedan}) = \frac{\text{black sedans}}{\text{total}} = \frac{1500}{5000} = 0.30, \quad \text{or along the path: } P(\text{black}) \times P(\text{sedan} \mid \text{black}) = 0.40 \times 0.75 = 0.30 = 30\%$$

(b) Number of white SUVs

$$\text{total} \times P(\text{white}) \times P(\text{SUV} \mid \text{white}) = 5000 \times 0.60 \times 0.20 = 600$$

Key takeaways

- Multiply **along** a branch; add **across** branches.
- **With replacement:** the branch probabilities repeat. **Without replacement:** update both the counts and the total after each draw.
- "At least one ..." is usually easiest as $1 - P(\text{none})$.
- Watch for "in that order" (one path) versus "in any order" (several paths).